11 Lecture 11 (February 7, 2025, est.): Abelian reconstruction and transporters

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For the first part of reconstruction, forget the tensor structure. Let $\mathcal{C}$ be an essentially small k-linear abelian category and

$$\omega : \mathcal{C} \longrightarrow \mathrm{Vec}_k$$

an exact faithful k-linear functor, with finite-dimensional values. We will construct a coalgebra B and an equivalence $\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B)$ compatible with ω.

Finite-dimensional algebras and coalgebras

If A is a finite-dimensional algebra, its dual $A^* = \mathrm{Hom}_k(A, k)$ is a coalgebra, with

$$\Delta_{A^*}(\phi)(a \otimes b) = \phi(ab), \qquad \varepsilon_{A^*}(\phi) = \phi(1).$$

A finite-dimensional left A-module M becomes a right A^*-comodule. For a basis a_i of A with dual basis a_i^*, its coaction is

$$m \longmapsto \sum_i a_i m \otimes a_i^*.$$

This is independent of the basis. Associativity of the action is coassociativity of the coaction, and the unit condition is the counit identity. Conversely, a coaction gives $a \cdot m = (\mathrm{id} \otimes \mathrm{ev}_a)\rho_M(m)$. These constructions are inverse and preserve underlying vector spaces.

Tensoring an object by a vector space

A k-linear abelian category admits an action of finite-dimensional vector spaces. For $V = k^n$, define $V \otimes X = X^{\oplus n}$ and let matrices act by the corresponding matrices of morphisms. Choosing bases extends the construction to all finite-dimensional V; changes of basis give canonical comparison isomorphisms. Equivalently, the object is characterized by

$$\mathrm{Hom}_{\mathcal{C}}(V \otimes X, Y) \simeq \mathrm{Hom}_k\big(V, \mathrm{Hom}_{\mathcal{C}}(X, Y)\big).$$

Here the tensor product is this vector-space action, not a monoidal structure on $\mathcal{C}$. There is also a natural isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(Y, V \otimes X) \simeq V \otimes_k \mathrm{Hom}_{\mathcal{C}}(Y, X).$$

Write

$$\underline{\mathrm{Hom}}(V, X) = V^* \otimes X.$$

A k-linear functor commutes with these constructions up to their natural identifications; in particular

$$\omega\big(\underline{\mathrm{Hom}}(V, X)\big) \simeq \mathrm{Hom}_k(V, \omega X).$$

11.1 Definition. For a subspace $W \subseteq V$ and a subobject $Y \subseteq X$, the *transporter* of W into Y is

$$(Y : W) = \ker\big(\underline{\mathrm{Hom}}(V, X) \longrightarrow \underline{\mathrm{Hom}}(W, X/Y)\big).$$

Exactness gives

$$\omega(Y : W) = \{f : V \to \omega X \mid f(W) \subseteq \omega Y\}.$$

Thus linear conditions about preserving subspaces can be expressed by subobjects in $\mathcal{C}$.

The object of endomorphisms preserving subobjects

Faithfulness implies $\omega X = 0 \Rightarrow X = 0$. If $Y \subsetneq X$, exactness gives $\dim \omega Y < \dim \omega X$. Hence every object of $\mathcal{C}$ has finite length, and every descending chain of subobjects terminates. In particular, intersections of arbitrarily many subobjects of a fixed object are already finite intersections. Exactness of ω then preserves these intersections. It also reflects inclusions between subobjects of a common object: if $\omega P \subseteq \omega Q$, apply ω to $P \cap Q \subseteq P$ to conclude $P \subseteq Q$.

Fix X, put $V = \omega X$ and $H_X = \underline{\mathrm{Hom}}(V, X)$. For each $n \geq 1$, let

$$d_n : H_X \longrightarrow \underline{\mathrm{Hom}}(V^{\oplus n}, X^{\oplus n})$$

be the diagonal map. Define

$$P_X = \bigcap_{n \geq 1,\, Y \subseteq X^{\oplus n}} d_n^{-1}(Y : \omega Y) \subseteq H_X.$$

The intersection exists by the preceding finite-length argument. Its image under ω is

$$A_X := \omega(P_X) = \{a \in \mathrm{End}_k(V) \mid a^{\oplus n}(\omega Y) \subseteq \omega Y \text{ for all } Y \subseteq X^{\oplus n},\ n \geq 1\}.$$

This is a finite-dimensional unital subalgebra: identities preserve all the subspaces, and preservation is closed under linear combinations and composition.

11.2 Lemma. *The object P_X is the smallest subobject $Q \subseteq H_X$ such that* $\mathrm{id}_V \in \omega Q \subseteq \mathrm{End}_k(V)$.

Proof. The identity belongs to A_X, so P_X is one such subobject. Conversely, suppose $Q \subseteq H_X$ and $\mathrm{id}_V \in \omega Q$. Choose a basis of V^* to identify H_X with $X^{\oplus d}$, $d = \dim V$. By the defining property of A_X, the diagonal action of every $a \in A_X$ preserves ωQ. Under $\omega H_X = \mathrm{End}_k(V)$, this diagonal action is left composition, $q \mapsto a \circ q$. Applying it to id_V shows $a \in \omega Q$. Thus $\omega P_X \subseteq \omega Q$, and reflection of subobject inclusions gives $P_X \subseteq Q$. ∎

Let $\langle X \rangle_{\oplus}$ denote the full subcategory consisting of subquotients of finite direct sums of X. This notation is different from the tensor-generated category $\mathcal{C}_X$ used

earlier. The category $\langle X\rangle_\oplus$ is abelian, and its inclusion in $\mathcal{C}$ is exact: kernels, cokernels and finite sums of such subquotients are again such subquotients.

Each $a \in A_X$ acts diagonally on $\omega(X^{\oplus n})$, preserves every subobject, and hence acts on the image under ω of every subquotient. These actions are independent of presentations and natural in $Y \in \langle X\rangle_\oplus$. To check this, take the graph of a morphism between two subquotients, pull it back to their presenting subobjects, and regard that pullback as a subobject of a direct sum of copies of X. Its preservation is exactly the assertion that the morphism intertwines the two actions. In particular, apply this to an identity morphism to compare presentations.

We have therefore lifted ω to an exact faithful functor

$$\widetilde{\omega}_X : \langle X\rangle_\oplus \longrightarrow A_X\text{-mod}_{\mathrm{fd}}.$$

The next lecture proves that it is an equivalence and that $A_X = \mathrm{End}(\omega|_{\langle X\rangle_\oplus})$. The construction here follows the abelian part of [Mil12, Chapter X, Section 3], with the transporter and minimal-subobject arguments made explicit.

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